

Incident Handler's Journal

Instructions & Objective: This journal serves as a comprehensive record of findings, investigation logs, tool takeaways, and key cybersecurity insights accumulated across hands-on activities and incident response exercises.

Journal Entry #1: Documenting a Cybersecurity Incident

Field	Details
Date & Entry #	July 30, 2026 Entry #1
Description	Documenting a Ransomware Incident Across Response Phases: ● Detection & Analysis: The organization initially detected a ransomware infection. External technical assistance was requested to perform deep technical analysis. ● Containment, Eradication & Recovery: Immediate containment steps included shutting down impacted computer systems. Due to operational limits, external specialized organizations were engaged to support full eradication and recovery.
Tool(s) Used	None specified in this phase.
The 5 W's	● Who: An organized group of unethical threat actors / hackers. ● What: A major ransomware security incident. ● Where: A healthcare organization's enterprise network. ● When: Tuesday at 9:00 AM. ● Why: Attackers leveraged a phishing campaign to gain initial access, subsequently deploying ransomware that encrypted critical files. The primary motive appears to be financial gain via ransom demand.
Additional Notes & Reflection Questions	● How can the healthcare provider strengthen defense-in-depth measures to prevent future phishing breaches? ● What are the legal, ethical, and operational implications regarding payment of ransom demands to recover decryption keys?

Journal Entry #2: Analyzing a Packet Capture File

Field	Details
Date & Entry #	July 30, 2026 Entry #2
Description	Analyzing network activity through a packet capture (PCAP) file to identify anomalies and potential malicious traffic patterns.
Tool(s) Used	Wireshark: A graphical network protocol analyzer that allows analysts to inspect packet headers, inspect protocol behavior, and investigate suspicious network communications in real time or via offline PCAP files.
The 5 W's	N/A (Tool exploration exercise).
Additional Notes	First-time exposure to Wireshark. While the complex GUI can feel overwhelming initially, its capability to break down protocol layers makes it an indispensable asset for network forensics.

Journal Entry #3: Capturing My First Packet

Field	Details
Date & Entry #	July 31, 2026 Entry #3
Description	Utilizing command-line utilities to perform live packet capture and apply filtering mechanisms on live network interfaces.
Tool(s) Used	tcpdump: A powerful command-line packet analyzer that enables security professionals to intercept, filter, and review TCP/IP traffic passing through network interfaces.
The 5 W's	N/A (Tool exploration exercise).

Field	Details
Additional Notes	As a CLI beginner, mastering syntax flags and filter flags presented a steep learning curve. Working through syntax errors systematically reinforced the importance of precision in command-line tools.

Journal Entry #4: Investigating a Suspicious File Hash

Field	Details
Date & Entry #	July 31, 2026 Entry #4
Description	Investigating Malicious File Hash (Detection & Analysis Phase): Acting as a SOC analyst investigating a security alert triggered by an Intrusion Detection System (IDS) to evaluate whether a flagged file hash corresponds to an active malware threat.
Tool(s) Used	VirusTotal: Threat intelligence platform that analyzes file hashes, domain names, IP addresses, and URLs against dozens of antivirus engines and community databases to identify Indicators of Compromise (IoCs).
The 5 W's	● Who: An unknown external threat actor. ● What: Phishing email containing a malicious attachment with SHA-256 hash: 54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93ba b527f6b ● Where: An employee endpoint at a financial services firm. ● When: 1:20 PM (SOC alert triggered upon IDS detection). ● Why: An employee downloaded and executed the malicious attachment from an unverified email message.
Additional Notes & Mitigation	● What controls can be implemented to block malicious email attachments prior to user execution? ● Is enhanced Security Awareness Training required to educate workforce members on identifying suspicious email indicators?

Course Reflections & Takeaways

Reflection Area	Key Insights
Challenging Activities	The tcpdump CLI activity was the most challenging due to unfamiliar syntax and parameters. Troubleshooting error outputs required patience, methodical instruction reviewing, and repeated execution, instilling a key practical lesson in operational diligence.
Evolution of Incident Response Understanding	Gained a comprehensive understanding of the full incident response lifecycle. Moving beyond basic definitions, the course highlighted the essential synergy between response frameworks, specialized investigation tools, structured playbooks, and cross-functional human teams.
Favorite Concept & Toolset	Network traffic analysis using packet protocol analyzers (Wireshark & tcpdump) was the most engaging topic. Inspecting live packet flows provided immediate, actionable insight into network behaviors and security postures.